

Symmetric rate perturbations of Hodgkin–Huxley gating are observationally equivalent: an identifiability constraint on mechanosensitivity experiments

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Abstract

Mechanical modulation of voltage-gated channel kinetics is usually represented in Hodgkin–Huxley models by multiplying the forward and reverse rates of a gate by a common strain-dependent factor. We show that this representation carries strictly less information than it appears to. For an arbitrary compartmental Hodgkin–Huxley system, the solution operator depends on the perturbation parameters only through the vector of rate multipliers, so every pair of mechanistic stories that induces the same multipliers defines the identical dynamical system, and hence the identical voltage trajectory, under every stimulus and every initial condition. Three consequences follow. First, the coupling coefficient and the imposed strain are not separately identifiable from a single strain level; only their product is. Second, a spatially and gate-uniform strain perturbation is indistinguishable from the kinetic component of a temperature change: a reported 1.4-fold acceleration is reproduced exactly by a rise of 3.06°C at $Q_{10} = 3$. Third, no measurement of the membrane-voltage trajectory, however detailed, can separate them, so agreement between a strain-coupled model and a recorded spike waveform is not evidence of mechanical coupling. We then identify what does discriminate. The gate-selective component of the log-multiplier vector is identifiable and is not produced by uniform thermal scaling; the symmetric-scaling hypothesis predicts a voltage-independent log ratio of relaxation times together with an unshifted steady-state activation curve, which separates it from effective-voltage coupling; and areal strain and warming act on channel surface density with opposite sign, predicting a divergence of 8 to 24% in peak conductance at the reference perturbation whose sign does not depend

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on the poorly constrained conductance Q_{10} . We give the resulting design constraints, including a temperature-control tolerance of 0.31°C if thermal drift is to account for no more than a tenth of a reported 1.4-fold effect.

Keywords: structural identifiability, Hodgkin–Huxley, mechanosensitivity, observational equivalence, Q_{10} , experimental design

1. Introduction

Voltage-gated sodium channels respond to membrane mechanics. Stretch reversibly accelerates both activation and inactivation of sodium current [12], and channel behaviour more generally depends on membrane tension, curvature and lipid environment [6, 13, 15]. The Hodgkin–Huxley (HH) formalism [10] has no mechanically defined state variable, so representing these effects requires an added constitutive term. The usual choice, motivated by transition-state theory [8], is to suppose that strain lowers a kinetic barrier and therefore multiplies the forward and reverse rates of a gate by a common factor $\exp(q_x\varepsilon)$, where ε is fractional areal strain and q_x is a coupling coefficient. The construction is attractive because it preserves the steady-state activation curve exactly and changes only relaxation times, which appears to isolate a purely kinetic effect.

This note asks a prior question: what can such a model be evidence for? The answer is more restrictive than the modelling literature generally assumes, and the restriction is structural rather than statistical. It does not depend on measurement noise, on the amount of data, or on the fitting procedure.

The argument is an application of structural identifiability analysis [1, 5, 4] to the particular reparameterisation that symmetric rate scaling induces. The identifiability of Hodgkin–Huxley parameters has been examined before for the unperturbed model: for a generalised HH system fitted to voltage-clamp data, the steady-state gating parameters are not separately identifiable while the relaxation time constants are [16]. Their question is complementary to the one asked here. They ask what can be recovered about an HH system from clamp data; we take the unperturbed system as given and ask what can be recovered about a perturbation applied to it. Their conclusion that time constants are identifiable is what makes the multiplier vector identifiable below, and the degeneracy reported here is therefore not a restatement of theirs but a second one, sitting one level up. The central observation is elementary once stated: the vector field of a compartmental HH system depends on the perturbation parameters only through a finite vector

of positive multipliers, one per gate per compartment. Everything that the model can express about the perturbation is contained in that vector. Any two physically distinct hypotheses that agree on it are indistinguishable by any experiment that observes the state trajectory, including one with perfect data.

This has a sharp corollary that we have not seen stated. Kinetic Q_{10} scaling of the rate functions [9] is a multiplicative perturbation of exactly the same class. A uniform strain perturbation and a matched temperature change are therefore not merely hard to distinguish from spike-shape data; they are formally the same model. Reporting that a strain-coupled model reproduces a recorded waveform is, to that extent, uninformative about mechanics.

The purpose of the note is not sceptical. Identifying the exact degeneracy identifies its complement, and the complement is a short and checkable list of measurements that do carry mechanical information. We state the degeneracy (Section 3), the discriminating signatures (Section 4), a numerical illustration in a two-compartment soma-axon model (Section 5), and the resulting design constraints (Section 6).

2. Setup

2.1. Compartmental Hodgkin–Huxley systems

Let $j \in \{1, \dots, N\}$ index compartments and let $\mathcal{G}$ be a finite set of gating variables, with x_j the value of gate $x \in \mathcal{G}$ in compartment j . Write V_j for membrane voltage. A compartmental HH system has the form

$$C_j \frac{dV_j}{dt} = - \sum_i \bar{g}_{i,j} \Pi_i(\{x_j\}_{x \in \mathcal{G}}) (V_j - E_i) + \sum_k g_{jk} (V_k - V_j) + I_j(t), \quad (1)$$

$$\frac{dx_j}{dt} = \alpha_x^{(j)}(V_j) (1 - x_j) - \beta_x^{(j)}(V_j) x_j, \quad x \in \mathcal{G}, \quad (2)$$

where Π_i is a monomial in the gating variables (for instance $m_j^3 h_j$ for the sodium current), $\bar{g}_{i,j}$ are conductance densities, E_i reversal potentials, g_{jk} axial conductances and I_j an applied current. Nothing below depends on N , on the number of currents, on the monomials Π_i , or on the topology of the axial coupling.

2.2. The multiplicative perturbation class

Let α_x^0, β_x^0 denote unperturbed rate functions. We consider perturbations in the class

$$\mathcal{M}: \quad \alpha_x^{(j)}(V) = \rho_{x,j} \alpha_x^0(V), \quad \beta_x^{(j)}(V) = \rho_{x,j} \beta_x^0(V), \quad \rho_{x,j} > 0. \quad (3)$$

A perturbation hypothesis is a parameterisation $\theta \mapsto \rho(\theta)$ into $\mathbb{R}_{>0}^{\mathcal{G} \times N}$. Three instances recur below.

Strain coupling. With ε_j the fractional areal strain $(A_j - A_{j,0})/A_{j,0}$ in compartment j and $q_x = \chi_x/k_B T$ the barrier sensitivity of gate x , transition-state theory gives

$$\rho_{x,j}^{\text{strain}} = \exp(q_x \varepsilon_j). \quad (4)$$

Kinetic temperature scaling. With ΔT_j a temperature offset and Q_{10} the usual coefficient,

$$\rho_{x,j}^{\text{temp}} = Q_{10}^{\Delta T_j/10}. \quad (5)$$

Isoform or modulatory rate scaling. Any hypothesis asserting that a gate runs faster by a fixed factor in some compartments, without change of voltage dependence, also lies in $\mathcal{M}$.

We write $\eta_{x,j} = \log \rho_{x,j}$ throughout.

3. Observational equivalence

Proposition 1 (Invariance and scaling). *Under (3), for every gate x , compartment j and voltage V ,*

$$\begin{aligned} x_\infty^{(j)}(V) &= \frac{\rho_{x,j} \alpha_x^0}{\rho_{x,j} \alpha_x^0 + \rho_{x,j} \beta_x^0} = x_\infty^0(V), \\ \tau_x^{(j)}(V) &= \frac{1}{\rho_{x,j} (\alpha_x^0 + \beta_x^0)} = \rho_{x,j}^{-1} \tau_x^0(V). \end{aligned} \quad (6)$$

The steady-state occupancy is unchanged at every voltage and the relaxation time is uniformly rescaled. This is the property that makes $\mathcal{M}$ attractive as a minimal coupling, and it is also, as the next result shows, the source of its degeneracy.

Proposition 2 (Factorisation of the solution operator). *Let $S(\theta; I, z_0)$ denote the solution of (1)–(2) under hypothesis θ with applied current I and initial condition z_0 . Then S factors through the multiplier vector: there is a map $\tilde{S}$ with*

$$S(\theta; I, z_0) = \tilde{S}(\rho(\theta); I, z_0). \quad (7)$$

Consequently, if $\rho(\theta) = \rho(\theta')$ then θ and θ' produce identical trajectories for every applied current, every initial condition and every choice of the remaining electrical and structural parameters, and hence identical values of every functional of the state trajectory.

Proof. The right-hand side of (1) does not involve θ . The right-hand side of (2) involves θ only through the products $\rho_{x,j}\alpha_x^0$ and $\rho_{x,j}\beta_x^0$. Two hypotheses with the same ρ therefore define the same vector field on the same state space, and the solutions of an initial value problem with identical vector field, identical input and identical initial data coincide. $\square$

Proposition 2 is trivial to prove and consequential to apply. The fibres of ρ are exact indistinguishability classes. No experiment that observes voltage, current, or gating occupancy as a function of time can distinguish two hypotheses lying in the same fibre, irrespective of data volume or noise level. In the terminology of structural identifiability [1, 4], ρ is the maximal identifiable function of θ within $\mathcal{M}$; ρ itself is identifiable, since by (6) a single relaxation-time measurement at one voltage determines $\rho_{x,j}$ given τ_x^0 .

Corollary 3 (Coefficient and strain are not separately identifiable). *Under (4) with a single imposed strain ε , the pairs (q_x, ε) and $(\lambda q_x, \varepsilon/\lambda)$ are indistinguishable for every $\lambda > 0$. Only the product $\eta_x = q_x \varepsilon$ is identifiable.*

The coefficient q_x becomes identifiable only if ε is measured independently, for instance from patch geometry, or if two or more strain levels are imposed and the linearity of η_x in ε is itself tested rather than assumed.

Corollary 4 (Strain and temperature are formally the same perturbation). *Strain coupling (4) and kinetic temperature scaling (5) are indistinguishable whenever*

$$q_x \varepsilon_j = \frac{\Delta T_j}{10} \log Q_{10} \quad \text{for all } x, j. \quad (8)$$

In particular, the 1.4-fold acceleration of sodium activation and inactivation reported under stretch [12] is reproduced exactly by

$$\Delta T = \frac{10 \log 1.4}{\log Q_{10}} = 3.06^\circ\text{C} \quad (Q_{10} = 3), \quad 3.67^\circ\text{C} \quad (Q_{10} = 2.5). \quad (9)$$

The practical force of Corollary 4 is that it applies at the level of the entire trajectory. It is not a statement that the two are hard to separate in noisy waveform data. They are, within the multiplicative class, the same model. It follows immediately that fitting a strain-coupled compartmental model to a recorded action potential, however well the fit succeeds, supplies no evidence that the perturbation is mechanical rather than thermal, and equally none that it is thermal rather than mechanical. The same holds for any whole-cell endpoint computed from the trajectory: onset rapidity, phase-plot slope, latency, threshold, or the relative timing of two compartments.

4. What does discriminate

Proposition 2 bounds the degeneracy exactly, so its complement is informative. Three escapes exist, and each corresponds to a distinct measurement.

4.1. Gate selectivity

Write the log-multiplier vector in a compartment as a common-mode part and a gate-selective part,

$$\eta_{x,j} = \bar{\eta}_j + \delta_{x,j}, \quad \bar{\eta}_j = \frac{1}{|\mathcal{G}|} \sum_{x \in \mathcal{G}} \eta_{x,j}, \quad \sum_{x \in \mathcal{G}} \delta_{x,j} = 0. \quad (10)$$

Proposition 5 (The gate-selective component is non-thermal). *Kinetic temperature scaling with a single Q_{10} lies on the diagonal $\eta_{m,j} = \eta_{h,j} = \eta_{n,j}$, so it contributes only to $\bar{\eta}_j$ and has $\delta_{x,j} = 0$. Any hypothesis with $\delta_{x,j} \neq 0$ is therefore not reproducible by uniform thermal scaling, and $\delta_{x,j}$ is identifiable by Proposition 2. Real thermal scaling is not uniform; Corollary 6 bounds the resulting departure from the diagonal.*

This is the most useful positive consequence of the analysis. It says that the informative quantity in a mechanosensitivity experiment is the *ratio* of accelerations across gates, not the absolute acceleration of any one of them. Reporting that stretch accelerated sodium activation by a factor ρ_m is consistent with a thermal artefact; reporting ρ_m/ρ_h , or ρ_m/ρ_n , is not. A single-gate experiment cannot in principle settle the question; a two-gate experiment can.

The caveat is that gate-specific Q_{10} values differ, so the thermal diagonal is an idealisation rather than a fact. The measured values are well separated: Q_{10} is 1.71 for sodium activation, 2.35 for sodium inactivation and 3.12 for potassium activation [9]. A real temperature change therefore has a non-zero gate-selective component, of size $|\log(Q_{10,m}/Q_{10,h})|/10 = 0.032$ per degree for the m - h pair and 0.060 per degree for the m - n pair.

The consequence is quantitative rather than fatal, and it is worth stating in both directions. An uncontrolled drift of 3.06°C , which is the drift that would by itself account for the whole of a 1.4-fold acceleration, yields $\rho_m/\rho_h = 0.907$: a gate ratio of the same order as the effects a mechanosensitivity study is trying to report, and hence a genuine confound if temperature is left to wander. Under the tolerance derived in Section 6, 0.31°C , the same artefact falls to 1.0% in ρ_m/ρ_h and 1.9% in ρ_m/ρ_n . The practically relevant claim is therefore a threshold rather than a dichotomy, and it is worth stating separately.

Corollary 6 (Thermal bound on the gate-selective component). *Suppose temperature is held within $\pm\Delta T_{\max}$ of its nominal value and that gate x has kinetic coefficient $Q_{10,x}$. Then the contribution of thermal variation to the gate-selective difference $\eta_x - \eta_y$ is bounded by*

$$|\eta_x - \eta_y|_{\text{thermal}} \leq \frac{\Delta T_{\max}}{10} \left| \log \frac{Q_{10,x}}{Q_{10,y}} \right|. \quad (11)$$

A measured $|\log(\rho_x/\rho_y)|$ exceeding this bound cannot be produced by temperature at the achieved tolerance, whatever the value of the common-mode acceleration.

Proof. Under (5) with offset ΔT , $\eta_x = (\Delta T/10) \log Q_{10,x}$, so $\eta_x - \eta_y = (\Delta T/10) \log(Q_{10,x}/Q_{10,y})$, which is monotone in ΔT and attains its extreme values at $\pm\Delta T_{\max}$. $\square$

Equation (11) is the form in which Proposition 5 should be used. It converts the qualitative claim that the gate-selective component is non-thermal into a number the experiment can be checked against, and it has the convenient consequence that the tolerance needed to control the common-mode confound controls the gate-selective one as well.

4.2. Separation from effective-voltage coupling

A competing constitutive hypothesis supposes that strain acts as an effective voltage offset, $\alpha_x(V + \zeta_x \varepsilon)$ and $\beta_x(V + \zeta_x \varepsilon)$. Write $\delta_x = \zeta_x \varepsilon$. This lies outside $\mathcal{M}$, and Proposition 1 fails for it.

Proposition 7 (Discriminating signature). *Define the relaxation-time log ratio*

$$L_x(V) = \log \frac{\tau_x(V, 0)}{\tau_x(V, \varepsilon)}. \quad (12)$$

Under symmetric scaling, $L_x(V) = \eta_x$ is constant in V , identical for all gates sharing a coefficient, and x_∞ is unchanged. Under effective-voltage coupling, $L_x(V) = \log [\tau_x^0(V)/\tau_x^0(V + \delta_x)]$, which is non-constant in V wherever τ_x^0 is not exponential, is gate-dependent through the shape of τ_x^0 , and is accompanied by a translation $x_\infty(V) \mapsto x_\infty^0(V + \delta_x)$.

The two predictions are independent and both are sharp. Neither requires a spike. Both require only a family of voltage-clamp steps repeated under at least two mechanical conditions, with the time constants and steady-state curves fitted jointly.

4.3. Channel surface density

A point that the multiplicative framing obscures deserves separate statement. Fractional areal strain by definition increases membrane area by a factor $(1 + \varepsilon)$. If the number of channels is conserved on the timescale of the perturbation, the surface density, and hence the conductance density appearing in (1), must fall:

$$\bar{g}_{i,j} \mapsto \frac{\bar{g}_{i,j}}{1 + \varepsilon_j}. \quad (13)$$

A temperature change acts on $\bar{g}$ in the opposite direction. Open-channel conductance carries the weak temperature dependence characteristic of a diffusion-limited process rather than the strong dependence of gating: the peak sodium current of rat myelinated fibre changes with Q_{10} near 1.1 between 40 and 20 °C and near 1.3 between 20 and 10 °C [14], and in a structurally unrelated channel, ClC-1, current amplitude gives $Q_{10} \approx 1.6$ against a gating Q_{10} of 3 to 4 [2]. The sodium-specific evidence is the first of these, and it gives 1.1 to 1.3; the ClC-1 figure is included not because chloride channels bear on Nav gating but because it is the upper end of what open-channel conduction is observed to do, and taking it as the upper bound makes the calculation deliberately unfavourable to the discriminator. On the band 1.1 to 1.7 so constructed, the reference perturbation $\varepsilon = 0.05$ with matched $\Delta T = 3.06$ °C gives the divergences of Table 1.

Table 1: Predicted change in peak conductance density at the reference perturbation, $\varepsilon = 0.05$ and matched $\Delta T = 3.06$ °C, across the plausible range of the conductance Q_{10} . The magnitude of the discriminator depends on a coefficient that is not tightly constrained; its sign does not.

	conductance Q_{10}	strain	warming	divergence
lower bound	1.1	$0.952 \bar{g}$	$1.030 \bar{g}$	8.1%
reference	1.3	$0.952 \bar{g}$	$1.084 \bar{g}$	13.8%
upper bound	1.7	$0.952 \bar{g}$	$1.176 \bar{g}$	23.5%

Strain predicts the same dilution at every value, warming predicts an increase at every value, and the two therefore separate by between 8% and 24% with opposite signs throughout. This is measurable in the same voltage-clamp family that yields the kinetics, and it costs nothing additional to record. Equation (13) is not an optional refinement: a model that imposes areal strain on gating rates while holding conductance density fixed is internally inconsistent, and it discards one of the three available discriminators.

5. Numerical illustration

5.1. Model

We use a two-compartment soma–axon system, $j \in \{s, a\}$, with classical squid rate functions [10] at their original benchmark temperature, sodium conductance concentrated in the axonal compartment, and resistive axial coupling. Parameters are in Table 2. Both compartments are initialised at the fixed point of the unstimulated system, obtained by Newton solve, rather than at a nominal resting voltage; the two differ, and the difference is large enough at high sodium ratios to produce a spurious spike during the startup transient. Equations are integrated by fixed-step fourth-order Runge–Kutta with $\Delta t = 0.002$ ms.

Table 2: Parameters of the illustrative two-compartment system. The perturbation is applied to the axonal compartment only.

C_m	$1.0 \mu\text{F}/\text{cm}^2$	E_{Na}	50 mV
$\bar{g}_{\text{Na},s}$	$120 \text{ mS}/\text{cm}^2$	E_K	−77 mV
$\bar{g}_{\text{Na},a}$	$240 \text{ mS}/\text{cm}^2$	E_L	−54.387 mV
$\bar{g}_K$	$36 \text{ mS}/\text{cm}^2$	g_c	$2.0 \text{ mS}/\text{cm}^2$
g_L	$0.3 \text{ mS}/\text{cm}^2$	I_{ext}	$20 \mu\text{A}/\text{cm}^2$, 5–6 ms

5.2. Three stories, one trajectory

Figure 1 overlays the somatic voltage under three perturbations that are physically distinct and numerically identical: areal strain of $\varepsilon = 0.05$ with $q_x = 6.73$; areal strain of $\varepsilon = 0.01$ with $q_x = 33.65$; and a purely kinetic temperature rise of 3.06°C at $Q_{10} = 3$. All three yield $\rho_m = \rho_h = 1.4$ in the axonal compartment, and the three trajectories agree to the last bit of double precision, with a maximum absolute difference of 0 mV. The agreement is exact rather than merely close because the three parameterisations collapse to the same multiplier vector before the vector field is evaluated, so the integrator performs an identical sequence of floating-point operations in all three runs. Bit-level agreement is a consequence of that construction; the result being claimed is the equality of the vector fields, which holds in exact arithmetic. The perturbation itself is not small: the maximum absolute deviation from the unperturbed control is 27.0 mV.

5.3. Weak waveform discrimination, strong clamp discrimination

Figure 2 contrasts symmetric scaling with the effective-voltage alternative. To avoid an unfair comparison we calibrate the effective-voltage shift

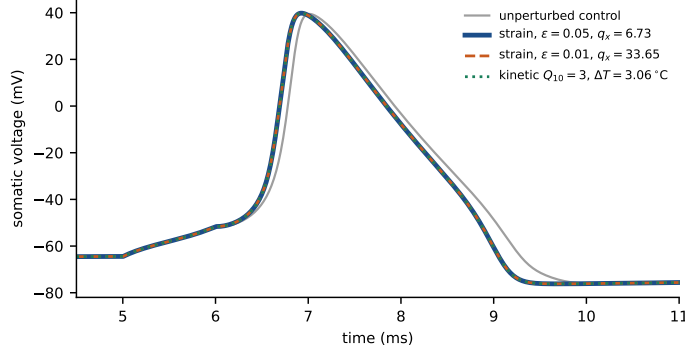


Figure 1: Observational equivalence within the multiplicative class. Somatic voltage under three distinct mechanistic hypotheses that induce the same axonal rate multipliers: reference strain ($\varepsilon = 0.05$, heavy solid line), reduced strain with higher barrier sensitivity ($\varepsilon = 0.01$, dashed line), and a purely kinetic temperature rise ($\Delta T = 3.06^\circ\text{C}$, dotted line). The three traces are numerically identical and overlaid one another; the light grey trace is the unperturbed control. The effect relative to control reaches 27.0 mV, so the coincidence is not an artefact of a small perturbation.

to reproduce the somatic spike timing of the symmetric model rather than matching a time constant at an arbitrary voltage; this gives $\delta = 2.29$ mV. The residual waveform difference is then 7.5 mV at maximum against an effect size of 27.0 mV relative to control, so whole-cell data distinguishes the two hypotheses, but weakly, and only after the structural and electrical parameters have been fixed independently.

Spike timing is chosen as the calibration target because it is the endpoint an experimenter fitting whole-cell data would actually match, but it is not the choice most favourable to the effective-voltage hypothesis, and the conclusion does not depend on it. Minimising the maximum waveform residual directly over δ gives $\delta = 2.79$ mV and a residual of 5.9 mV, that is 21.9% of the effect size rather than 27.8%. Waveform discrimination is therefore weaker under the best available calibration of the competitor than under the timing-matched one, so the timing-matched figure understates the degeneracy rather than overstating it. The clamp-level separation below survives both: at $\delta = 2.79$ mV, $L_m(V)$ ranges over $[-0.152, 0.066]$ and $L_h(V)$ over $[-0.105, 0.176]$, still changing sign and still nowhere near the flat 0.336 predicted by symmetric scaling.

The clamp-level signatures separate qualitatively. Under symmetric scaling $L_x(V)$ is flat at $\log 1.4 = 0.336$ for both gates; under the timing-matched effective-voltage model $L_m(V)$ ranges over $[-0.124, 0.054]$ and $L_h(V)$ over $[-0.087, 0.144]$, both changing sign within the physiological range and differ-

ing from each other in shape. The steady-state curves are unshifted in the first case and translated by 2.29 mV in the second.

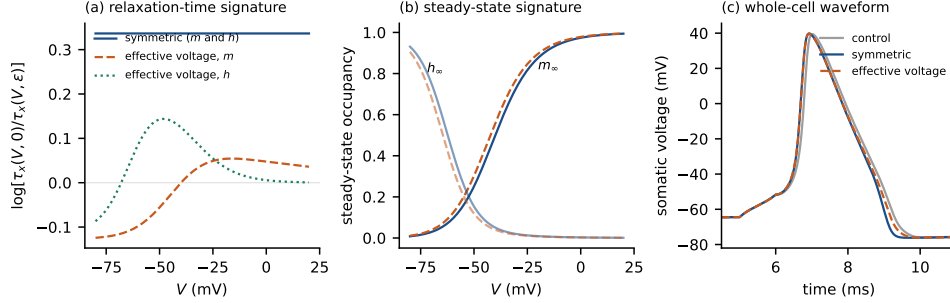


Figure 2: Discriminating symmetric scaling from effective-voltage coupling, with the effective-voltage shift calibrated to match somatic spike timing. (a) Relaxation-time log ratio $L_x(V)$: constant and gate-independent under symmetric scaling (solid line), voltage-dependent and gate-dependent under effective-voltage coupling (m dashed, h dotted). (b) Steady-state occupancies m_∞ and h_∞ : unchanged under symmetric scaling (solid lines), translated under effective-voltage coupling (dashed lines). (c) Somatic waveforms: unperturbed control (light grey), symmetric scaling (solid) and effective-voltage coupling (dashed). The two hypotheses nearly coincide while both depart clearly from control, illustrating that whole-cell data discriminates weakly where clamp data discriminates qualitatively.

5.4. Scope of the illustration

We deliberately do not report the effect of the perturbation on onset rapidity, compartment ordering, or threshold. By Proposition 2 such quantities are functionals of the trajectory, hence constant across the equivalence class, hence uninformative about the mechanical hypothesis. They characterise the consequences of localised kinetic acceleration, which is a different and much older question [3].

6. Consequences for experimental design

The analysis yields a short list of constraints. They are cheap and none requires apparatus beyond what a mechanosensitivity study already uses.

Clamp temperature to better than 0.31°C and report it.. By (9), an uncontrolled drift of ΔT accounts for a fraction $\Delta T \log Q_{10}/(10 \log \rho)$ of an observed acceleration ρ . If thermal drift is to explain no more than a tenth of a reported 1.4-fold effect at $Q_{10} = 3$, temperature must be held to 0.31°C . Mechanical stimulation by pressure or flow is not thermally neutral by default, and bath temperature at the headstage is not the relevant quantity.

Measure at least two gates and report their ratio.. By Proposition 5, the common-mode acceleration is confounded and the gate-selective component is not. A study reporting only sodium activation cannot distinguish its result from a uniform rate artefact. Reporting ρ_m/ρ_h against the spread of independently measured gate-specific thermal coefficients converts a confounded observation into an identifiable one.

Measure τ_x across voltage, not at one voltage.. By Proposition 7, the discriminating observable is the *constancy* of $L_x(V)$, which a single-voltage measurement cannot assess. A systematic voltage dependence of L_x , or a strain-dependent translation of x_∞ , rejects the symmetric barrier hypothesis and indicates coupling to the open–closed free-energy difference rather than to the barrier alone.

Record peak conductance.. By Section 4.3, strain and warming move $\bar{g}$ in opposite directions, with a predicted divergence of 8 to 24% at the reference perturbation depending on the conductance Q_{10} and with a sign that does not depend on it. This is the single cheapest discriminator available and it is usually already in the data.

Impose at least two strain levels, and measure strain.. By Corollary 3, one level identifies only η_x . Two or more levels permit the assumed linearity of η_x in ε to be tested rather than imposed. Independent estimation of ε from patch geometry is what converts η_x into the physical coefficient q_x .

Do not treat waveform agreement as evidence.. By Corollary 4, agreement between a strain-coupled compartmental model and a recorded action potential is compatible with a purely thermal perturbation of the same magnitude and carries no mechanical information.

Remark 8 (Magnitude of the implied coupling). The analysis says nothing about whether the reference coupling is physically reasonable, but the identifiable quantity makes that easy to check. Writing the barrier change as $\delta\Delta G_x^\ddagger = -\Delta\gamma \Delta A_x^\ddagger$ with $\Delta\gamma = K_A\varepsilon$, the implied in-plane area change at the transition state is $\Delta A_x^\ddagger = \eta_x k_B T / (K_A\varepsilon)$. For $\eta_x = \log 1.4$, $\varepsilon = 0.05$ and a bilayer area-expansion modulus $K_A \approx 0.24$ N/m [7], this is $\Delta A_x^\ddagger \approx 0.11$ nm², rising to roughly 0.45 nm² at a tension increment of 3 mN/m. These are modest values by comparison with mechanosensitive channel gating areas. The same calculation shows that $\varepsilon = 0.05$ at that modulus corresponds to a tension increment near 12 mN/m. Artificial bilayers and biological membranes stretch elastically only until their area rises by some two to three per

cent, rupturing thereafter at tensions of roughly 1 to 12 mN/m [11], so a bilayer cannot sustain $\varepsilon = 0.05$ at all, and ε in this context should be read as apparent whole-cell areal strain, which includes unfolding of membrane reservoirs, rather than as bilayer strain, of which such preparations carry a substantial reservoir. Whether a given ε is achievable in a given preparation is an experimental question, and the analysis is conditional on it, but the conditional is harmless: by Corollary 3 only the product $q_x\varepsilon$ is identifiable, so a smaller achievable strain rescales the implied coefficient without altering any degeneracy or any discriminator. The reference perturbation fixes a scale for the numbers quoted here; it is not a claim about what a membrane can sustain.

7. Scope and limitations

The equivalence result is confined to the multiplicative class $\mathcal{M}$. Couplings that are state-dependent, that act asymmetrically on forward and reverse rates, that alter the voltage dependence of the rate functions, or that introduce memory, all escape Proposition 2, and the corresponding models are correspondingly less degenerate. That is an argument for stating the coupling class explicitly rather than an argument against symmetric scaling.

The next class up is worth naming, because it is cheap to adopt and not degenerate in the same way. Allow the forward and reverse rates to carry separate multipliers, $\alpha_x \mapsto \rho_x^\alpha \alpha_x^0$ and $\beta_x \mapsto \rho_x^\beta \beta_x^0$. Proposition 1 then fails, since

$$x_\infty = \frac{\rho_x^\alpha \alpha_x^0}{\rho_x^\alpha \alpha_x^0 + \rho_x^\beta \beta_x^0},$$

which reduces to the unperturbed curve only when $\rho_x^\alpha = \rho_x^\beta$. The asymmetric part of the perturbation therefore appears in the steady-state activation curve, where it is directly measurable, while the symmetric part remains confounded with temperature exactly as before. Physically this is the difference between a coupling that moves the barrier alone and one that also moves the open–closed free-energy difference, and it is the model to fit when the steady-state curve is observed to shift.

The thermal equivalence of Corollary 4 is exact only for the kinetic component of a temperature change. Real temperature changes also scale maximal conductances and shift reversal potentials through the Nernst relation, and gate-specific Q_{10} values differ. Each of these breaks the degeneracy, and each is the basis of one of the discriminators in Section 6. The correct summary is that the kinetic and thermal hypotheses coincide exactly on the gating

equations and diverge only through effects the multiplicative framing omits, which is precisely why omitting them is costly.

Nothing here bears on whether voltage-gated channels are mechanosensitive. That is an experimental question with an affirmative answer in several preparations [12, 6, 13]. The claim is narrower: within the modelling idiom most commonly used to represent the effect, a large part of the apparent content is not identifiable, and the identifiable part is not the part usually reported.

8. Conclusion

Symmetric scaling of Hodgkin–Huxley gating rates is a well-motivated minimal coupling, and its two attractive properties, invariance of the steady-state curve and pure rescaling of relaxation time, are exactly the properties that make it observationally degenerate. The solution operator of an arbitrary compartmental system depends on the perturbation only through the vector of rate multipliers. Strain, temperature and isoform substitution can therefore be made to coincide exactly, and a 1.4-fold acceleration is equivalent to a 3.06 °C rise at $Q_{10} = 3$.

What survives is specific and testable. The gate-selective component of the log-multiplier vector is identifiable and is not produced by uniform thermal scaling. Symmetric scaling predicts a voltage-independent log ratio of relaxation times together with an unshifted steady-state curve, which separates it from effective-voltage coupling. Areal strain and warming act on channel surface density with opposite sign. A mechanosensitivity experiment that controls temperature tightly, measures two gates, varies voltage, records peak conductance, and imposes more than one strain level can decide the question. One that fits a spike waveform cannot.

Data and code availability

The simulation code and the scripts generating both figures and all reported numerical values are archived at Zenodo, <https://doi.org/10.5281/zenodo.22828618>, and are also available from the author.

Declaration of competing interest

The author declares no competing interests.

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